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Meteorological Conditions and Rural Injury Accidents in Iceland

Analysis of Accident Frequency and Traffic, 2007–2025

Ásgeir Snær Magnússon

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in Software Engineering

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Analysis of Accident Frequency and Traffic, 2007–2025

Ásgeir Snær Magnússon

30 ECTS thesis submitted in partial fulfillment of a
Master of Science degree in Software Engineering

Supervisors
Kristján Jónasson
Guðmundur Freyr Úlfarsson

Faculty of Industrial Engineering, Mechanical Engineering and Computer Science
School of Engineering and Natural Sciences
University of Iceland
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Faculty of Industrial Engineering, Mechanical Engineering and Computer Science
School of Engineering and Natural Sciences
University of Iceland
Dunhagi 5
107, Reykjavik, Iceland
Telephone: 525 4000

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Abstract

This study examines how rural injury-accident occurrence varies with weather conditions in Iceland from 2007–2025. The main focus is on mean wind speed, with additional analyses of wind gusts, temperature, season, time of day, and daylight. Accident records were linked to 10-minute weather observations from nearby stations. The number of accidents under different weather conditions was then compared with how often those conditions occurred locally. A matched-time comparison assessed whether the pattern persisted when recurring calendar conditions were held fixed. Traffic data were used in supporting analyses.

A total of 6,259 injury accidents were matched to a valid wind observation within 20 km and five minutes of the accident time. At mean wind speeds of 20 m/s or more, 68 injury accidents were observed, compared with 31.6 expected from the local frequency of such winds. This gives an observed-to-expected ratio of 2.15. For serious or fatal accidents, the corresponding ratio was 1.87, based on 12 observed accidents and 6.4 expected. In the matched-time comparison, the odds ratio for mean wind of at least 15 m/s versus 0–5 m/s was 1.61 (95% confidence interval 1.38–1.87). High wind gusts showed a similar pattern. Analyses using traffic data pointed in the same direction, although they were based on smaller samples. The results for temperature were less consistent.

Overall, rural injury accidents occurred more often during strong wind than would be expected from how often strong wind occurred locally. There was no clear evidence that accident severity differed between the wind categories.

These results show statistical associations, not causal effects. Because hourly traffic data are not available at the required spatial and temporal resolution, precise accident risk per vehicle cannot be estimated for the full study period.

Ágrip

Rannsóknin kannar hvernig tíðni umferðarslysa með meiðslum í dreifbýli á Íslandi breytist eftir veðurskilyrðum á árunum 2007–2025. Megináhersla er á meðalvindhraða, en einnig eru skoðaðar vindhviður, hitastig, árstíðir, tími dags og dagsbirta. Slysaögn voru tengd við tíu mínútna veðurmælingar frá nálægum veðurstöðvum. Fjöldi slysa við tiltekin veðurskilyrði var borinn saman við væntan fjölda miðað við tíðni skilyrðanna á sama stað og árstíma. Til stuðnings var veður á slysatíma borið saman við veður á sambærilegum tímum án slyss. Umferðargögn voru notuð í viðbótargreiningum til að varpa ljósi á túlkun niðurstaðnanna.

Alls voru 6.259 slys með meiðslum tengd við gilda vindmælingu frá veðurstöð í allt að 20 km fjarlægð og innan fimm mínútna frá slysatíma. Við meðalvindhraða 20 m/s eða meiri voru skráð 68 slys, en væntur fjöldi miðað við staðbundna tíðni slíkra vindskilyrða var 31,6. Hlutfall skráðs og vænts fjölda slysa var því 2,15. Fyrir alvarleg slys og banaslys var hlutfallið 1,87, eða 12 skráð slys á móti 6,4 væntum. Í pörðum samanburði slysatíma og sambærilegra tíma án slyss var gagnlíkindahlutfallið 1,61 (95% öryggisbil 1,38–1,87) fyrir meðalvindhraða að minnsta kosti 15 m/s, samanborið við 0–5 m/s. Sambærilegt mynstur kom fram við miklar vindhviður. Greiningar með umferðargögnum bentu í sömu átt, en byggðust á minni úrtökum. Tengsl hitastigs og slysa voru breytileg eftir hitabili og greiningaraðferð.

Slys með meiðslum voru hlutfallslega tíðari í miklum vindi en búast mætti við út frá staðbundinni tíðni hans. Ekki komu þó fram skýrar vísbendingar um meiri alvarleika meðal skráðra slysa með meiðslum í miklum vindi. Niðurstöðurnar lýsa tölfræðilegum tengslum en sýna ekki fram á orsakasamband. Umferðargögn á klukkustund liggja ekki fyrir með nægilegri upplausn í tíma og rúmi til að meta nákvæma slysaáhættu á hvern ekinn kílómetra við einstök veðurskilyrði fyrir allt rannsóknartímabilið.

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Abbreviations

ADU	Average annual daily traffic.
CI	Confidence interval.
O/E	Observed accidents divided by expected accidents.
OR	Odds ratio.
RR	Rate ratio.
SDU	Summer daily traffic, June–September.
VDU	Winter daily traffic, December–March.
VHDU	Spring/autumn daily traffic derived from ADU, SDU and VDU.
VKT	Vehicle-kilometres.

1 Introduction

1.1 Context and Motivation

Weather affects both driving conditions and the decision to travel. Strong wind can make a vehicle harder to control, low temperature can coincide with slippery roads, and darkness can reduce visibility. These conditions are relevant to Icelandic roads, where terrain can produce substantial local variation in wind (Snæbjörnsson et al., 2007; Pu et al., 2018). Precipitation and fog are also relevant road-weather hazards, but they are context for this study and are not analysed here.

Previous research provides two reasons to examine wind. Engineering studies describe mechanisms by which crosswind can destabilise vehicles, while observational studies compare accident occurrence with the frequency of windy conditions (Baker, 1986, 1994; Edwards, 1994). These are complementary forms of evidence: a physical mechanism does not establish how often accidents occur across a road network. Reviews also find less consistent road-safety evidence for wind than for precipitation (Theofilatos and Yannis, 2014). Local observations are therefore needed to assess the pattern in the setting under study.

In Iceland, wind-related road safety has been studied through vehicle reliability and operational vehicle-wind models (Snæbjörnsson et al., 2007; Hrafnkelsdóttir, 2020). The question addressed here is different: how does recorded rural injury-accident occurrence vary with the weather that is observed locally? The contribution is to connect a long Icelandic accident series with station- and season-specific weather frequency, then assess the interpretation through matched-time and traffic comparisons. This links the engineering motivation to population-level accident records without treating station wind as a measurement of the force on a vehicle.

This thesis studies rural accidents involving injury in Iceland from 2007–2025. Accident times and locations are linked to nearby 10-minute weather observations. The main comparison concerns mean wind speed. Wind gusts, temperature, season, time of day, daylight, and the available traffic data are then examined separately.

Accident counts alone do not measure accident risk. Severe weather can increase the hazard faced by each vehicle while reducing the number of vehicles on the road. Weather frequency and traffic volume are therefore considered separately (Andrey et al., 2003; Lord and Mannering, 2010; Washington et al., 2020).

1.2 Aim and Research Questions

The aim is to quantify how the occurrence of rural injury accidents varies with weather and what the available traffic data add to the mean-wind result. Three questions guide the analysis:

1. **Q1 — Primary weather-frequency O/E:** How does accident occurrence relative to local weather frequency vary across mean-wind, wind-gust and temperature intervals, overall and by season? The figures distinguish minor injury from serious or fatal injury; the combined all-injury result provides the main comparison across methods.
2. **Q2 — Matched-time robustness comparison:** Is the weather-frequency pattern supported when each accident time is compared with matched non-accident times at the same station and recurring calendar time?
3. **Q3 — Supporting traffic/exposure analyses:** What do the available annual and daily traffic data add to the interpretation of the wind pattern, given their coverage and exposure-allocation assumptions?

The primary 2007–2025 analysis addresses the first question, the matched-time analysis the second, and the smaller traffic-linked samples the third. Minor-injury and serious-or-fatal accidents together form the all-injury outcome. The study estimates associations, not causal effects or a complete national accident rate per vehicle-kilometre.

1.3 Thesis Outline

Chapter 2 gives the background for the weather and traffic comparisons. Chapter 3 describes the data, data preparation, and statistical methods. Chapter 4 presents the results, beginning with mean wind. Chapter 5 discusses their interpretation and limitations, and Chapter 6 gives the conclusions. The appendix contains selected checks that support interpretation of the main results.

2 Background

2.1 Wind and Rural Road Accidents

Crosswind can produce overturning, lateral sideslip, or excessive yaw. The critical condition depends on vehicle geometry and mass, vehicle speed, road friction, wind direction, and driver response rather than wind speed alone (Baker, 1986, 1994). High-sided vehicles are particularly exposed, but the physical mechanisms also provide a reason to examine loss-of-control accidents more generally. These engineering studies establish physical plausibility and vehicle-specific failure mechanisms; they do not by themselves establish a population-level association between wind and accident occurrence.

This problem is directly relevant in Iceland, where terrain can accelerate and redirect near-surface wind over short distances. An Icelandic reliability study by Snæbjörnsson et al. (2007) modelled accident probability as the combined outcome of wind velocity and direction, road camber and friction, and vehicle speed. A station measures wind at one point, whereas vehicle stability also depends on the wind's direction relative to the road and on local road conditions. Terrain between a station and a road can alter the wind reaching the vehicle. Station observations therefore remain exposure proxies, and a shorter matching distance does not guarantee identical conditions. Mobile road measurement research similarly shows that roadside terrain creates marked spatial variation that a sparse fixed-station network cannot fully resolve (Pu et al., 2018).

Vegagerðin has developed a vehicle-wind model to support decisions about hazardous conditions and road closures, while also noting the limited experimental evidence available for passenger cars (Hrafnkelsdóttir, 2020). The present thesis does not test that operational model; it examines population-level accident occurrence using a separate data design.

Empirical work has compared the share of accidents occurring in high wind with the share of time that high wind was observed, an approach closely related to the observed/expected comparison used here (Edwards, 1994). Studies linking crashes to nearby weather stations likewise show that station observations can be informative while retaining measurement error (Young and Liesman, 2007). More broadly, Theofilatos and Yannis (2014) report less consistent road-safety evidence for wind than for precipitation.

2.2 Temperature, Daylight, and Time of Day

Temperature, daylight, and time of day add context that wind alone cannot provide. They affect road-surface conditions, visibility, travel patterns, and the composition of road users, but accident counts in these categories are not rates unless the corresponding amount of travel is known. Temperature effects can also be non-linear and season-specific: a Seoul time-series

study found its strongest cold-weather association below freezing in winter, while an hourly case-crossover study in Shenzhen found a J-shaped temperature relationship (Lee et al., 2014; Zhan et al., 2020). Temperature is therefore compared with its local seasonal frequency rather than treated as having one linear effect.

Darkness presents the same denominator problem. A method developed using Norwegian, Swedish, and Dutch data compared daylight and darkness within clock time and season, avoiding the interpretation of raw night-time counts as risk (Johansson et al., 2009). Here, raw hour, season, and daylight counts describe the sample. Matched daylight and adjusted severity models are used for numerical comparisons.

2.3 Traffic and Accident Rates

Weather can change the amount of traffic as well as the hazard. Research using automatic traffic recorders shows that extreme weather variables, including wind, can reduce or redistribute hourly traffic volume (Sathiaraj et al., 2018). Accident counts during strong wind can remain low even if risk to each travelling vehicle increases. High-resolution crash studies show the value of hourly weather and matched-time analyses rather than daily or monthly averages (Xing et al., 2019). The analyses distinguish three quantities: accident occurrence relative to weather frequency, rates based on estimated vehicle-kilometres, and changes in observed daily traffic. Studies with real-time traffic, weather, and road geometry show what a more complete analysis could achieve when such data are available (Ahmed et al., 2012).

3 Data and Methods

3.1 Data

3.1.1 Source Data

Table 3.1 summarises the source datasets. Official documentation for each source is listed in the bibliography (Samgöngustofa, 2026; Veðurstofa Íslands, 2026; Vegagerðin, 2026; Hagstofa Íslands, 2026).

Table 3.1: Source datasets. The study period is 2007–2025.

Data	Provider	Contents	Coverage
Accident	Samgöngustofa	Accident events and road links	126,607 events and 126,610 road links, 2007–2025
Weather	Icelandic Meteorological Office (IMO)	10-minute weather observations and station metadata	232,459,562 observations from 338 stations; 438 station records, 2007–2025
Traffic and roads	Vegagerðin	Annual road traffic, daily counter reports, counter locations, and road geometry	33,757 section-years, 2000–2025; 1,033,659 counter channel-days, 2019–2024
Urban areas	Statistics Iceland	Urban-area polygons	97 polygons, 2020–2024 publication

Records from the year 2025 are included in the accident, injury, vehicle, road-link, annual-traffic and weather datasets under the same selection and quality criteria as records from earlier years. Daily counter data are available through 2024. Table 3.2 defines the weather and traffic quantities used in the subsequent comparisons.

Table 3.2: Definitions used throughout the thesis

Term	Definition	Use in this thesis
f	10-minute mean wind speed, m/s	Main weather measure, matched within five minutes of the accident time.
fg	Wind gust reported in the same 10-minute observation, m/s	Additional weather measure; not a daily maximum.
t	Air temperature reported with each 10-minute observation, °C	Additional weather measure matched independently.
O/E	Observed accidents divided by accidents expected from local weather frequency	Main relative-occurrence result; traffic is not included.
ADU	Average vehicles per day over the year	Used to check and derive annual traffic quantities.
SDU	Average vehicles per day in June–September	Official summer traffic value.
VDU	Average vehicles per day in December–March	Official winter traffic value.
VH DU	Average vehicles per day in April–May and October–November	Spring/autumn daily traffic calculated as the residual from ADU, SDU and VDU.

3.1.2 Accident Data

The accident dataset contains 126,607 records from 2007–2025 (Samgöngustofa, 2026). One record lacks valid temporal or spatial information, leaving 126,606 valid records.

Rural status is derived from accident coordinates and Statistics Iceland urban-locality polygons (Hagstofa Íslands, 2026). Points inside an urban polygon are classified as Urban, and valid points outside all polygons as Rural. A defined Vestmannaeyjar area is also classified as Urban. The same 2020–2024 polygon dataset is applied to the full 2007–2025 accident period. Changes in urban boundaries before 2020 are not represented.

Applying this rural classification yields 24,005 accidents, of which 6,414 involve injury. Of these, 6,259 have a qualifying primary weather match within 20 km and five minutes.

Accident severity is recorded in `meidsli`: code 1 denotes a fatal accident, code 2 a serious accident, code 3 a minor-injury accident, and code 4 an accident without injury. The main outcome is $\text{meidsli} \leq 3$. The serious-or-fatal comparison uses $\text{meidsli} \leq 2$. The source data also contain 17,591 rural damage-only accidents; these are outside the injury outcome.

The O/E figures divide the injury outcome into two separate groups: blue for minor injury ($\text{meidsli}=3$), and red for serious or fatal injury ($\text{meidsli}=1$ or 2). An accident appears in only one group. Table 3.3 gives a worked example of each group in windy conditions. Expected counts are calculated separately for each group using its station–season accident totals and local weather frequencies (the full formula is given below). For minor injury, 56 observed accidents divided by 25.18 expected gives O/E 2.22; for serious or fatal injury, 12 divided by 6.41 gives O/E 1.87. These pooled examples illustrate the calculation, not the cause of any individual accident.

Table 3.3: Worked examples for the two O/E groups: mean wind at least 20 m/s, all years and seasons pooled.

Group	meidsli	Observed	Expected	O/E
Minor injury accidents	3	56	25.18	2.22
Severe/fatal accidents	1 or 2	12	6.41	1.87

Figure 3.1 shows the geographic distribution of the rural injury sample and highlights accidents matched to strong mean wind.

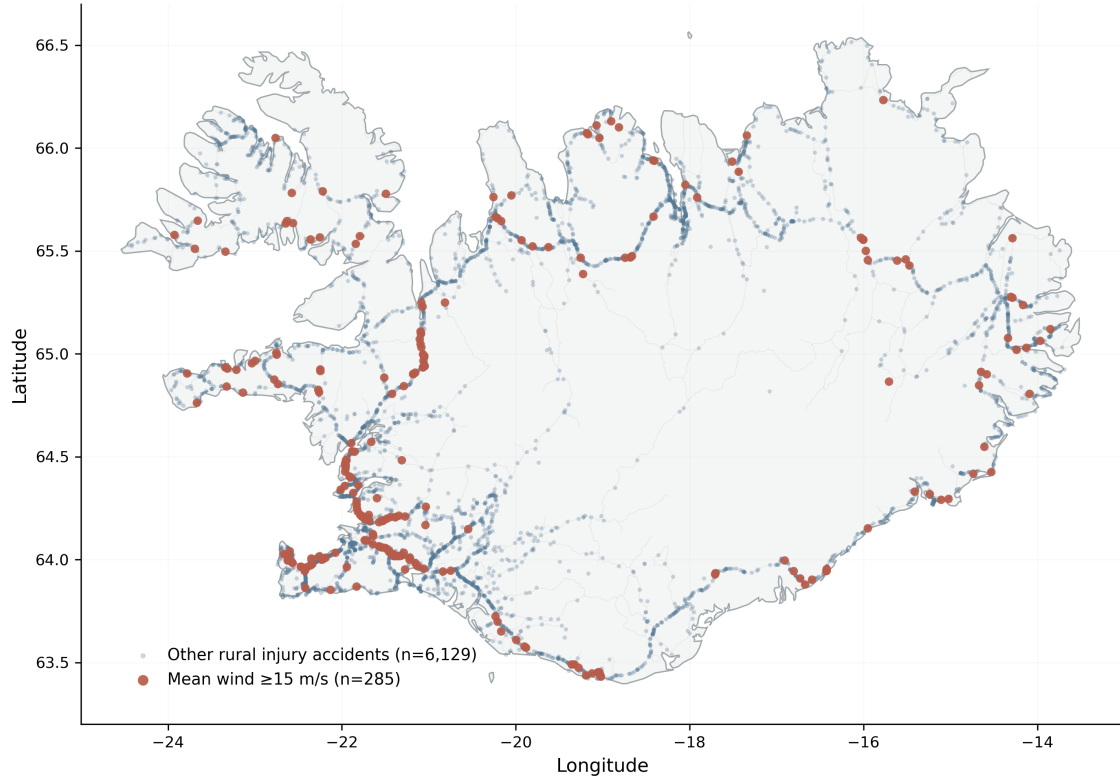


Figure 3.1: Locations of the 6,414 rural injury accidents. Accidents matched to mean wind of at least 15 m/s are highlighted. The map shows the spatial coverage of the study and is not a regional accident-rate map.

3.1.3 Weather Data

The weather dataset contains 10-minute observations from Icelandic weather stations. The variables used here are 10-minute mean wind speed (f), reported wind gust (fg), and temperature (t). For each accident, the gust is taken from the same matched 10-minute observation as mean wind; it is not the maximum across the accident day. Mean wind speed remains the main weather measure. A wind observation is used when

$$0 \leq f < 45, \quad 0 \leq fg < 65, \quad fg + 0.5 \geq f.$$

Rows missing either f or fg are excluded. Negative values, values at or above either upper limit, and internally inconsistent values where $fg = 0$ but $f > 0$ are excluded and reported in

the annual audit. A value of $f = 0$ alone is kept. Only uninterrupted all-zero runs where $f = fg = 0$ for at least two hours are excluded as potentially frozen sensors. Missing temperature does not affect whether a wind observation is used.

Each accident is matched to the geographically nearest station with a quality-controlled wind observation at one of the two adjacent timestamps on the 10-minute observation grid. The observations have 10-minute resolution, but the selected observation is no more than five minutes from the recorded accident time. The maximum matching distance is 20 km. The matching dataset includes all 6,414 accidents and records whether each accident meets the weather-match criterion.

Temperature is matched independently within the same 20 km and five-minute limits. The nearest valid value between -30 and 30°C may therefore come from a different station when the wind station lacks temperature. In the final analysis dataset, however, all 6,259 wind and temperature matches use the same station and observation time.

The IMO supplied observations of time, station, temperature, mean wind, and gust. The same quality-controlled weather data are used for accident matching, background frequencies, matched control times, and traffic analyses. No temperature value is imputed.

Table 3.4 reports spatial and temporal match quality for wind and temperature. Table 3.5 then summarises the wind-quality exclusions and retained observations.

Table 3.4: Share of accidents with weather information from a station with a valid measurement within 20 km. Distance is shown as median / P90; every retained observation is within five minutes of the accident time.

Variable	Accidents	Share	Stations	Distance (km)
Mean wind and gust	6,259	97.6%	279	5.1 / 12.3
Temperature	6,259	97.6%	279	5.1 / 12.3

Table 3.5: Wind-data scope and quality audit, 2007–2025. Quality-rule shares use all supplied observations. Every supplied station-year contains at least one wind measurement.

Category	Records	Share
All supplied station-time observations	232,459,562	100.00%
Missing f or fg	261,519	0.11%
Negative or upper-threshold wind	8,766	0.00%
Internally inconsistent f/fg	257,245	0.11%
Frozen all-zero runs (≥ 2 hours)	1,473,082	0.63%
Quality-controlled wind observations retained	230,458,950	99.14%

The wind-quality rules exclude 2,000,612 observations and retain 230,458,950 (99.14%). The largest excluded category is uninterrupted all-zero runs, which is reported separately because it reflects the rule for possible instrument stoppages rather than a missing observation. The audit reports each rule separately and uses all supplied weather observations from 2007–2025. For each accident’s station and season, the O/E expected count uses all quality-controlled background observations in that station–season group, not only observations recorded on accident days.

3.1.4 Traffic Data

Two traffic sources are used for different purposes.

Annual road-section data provide average annual daily traffic (ADU), summer daily traffic (SDU), winter daily traffic (VDU), section length, and annual vehicle-kilometres for 2000–2025 (Vegagerðin, 2026). The road-section comparison uses the years shared with the accident study, 2007–2025. Following the official definitions, SDU covers June–September and VDU covers December–March. These data can be linked to accidents with a registered road-section identifier, but exact matching is much narrower than the weather matching. Table 3.6 distinguishes the published averages from the derived spring/autumn value.

Table 3.6: Official traffic averages and the spring/autumn daily traffic value used in the annual-traffic analysis.

Period	Months	Daily traffic reference
ADU (<i>ársdagsumferð</i>)	Full calendar year	Official annual daily traffic
VDU (<i>vetrardagsumferð</i>)	December–March	Official winter daily traffic
SDU (<i>sumardagsumferð</i>)	June–September	Official summer daily traffic
VHDU (<i>vor- og haustdagsumferð</i>)	April–May and October–November	Residual daily average calculated from ADU, SDU, VDU, and the number of days in each period

Throughout the analyses, the four study seasons are aligned with these traffic periods: winter is December–March, spring is April–May, summer is June–September, and autumn is October–November. These are study-specific traffic-aligned seasons rather than conventional three-month meteorological seasons. Using June–September as summer keeps the weather comparison aligned with the Icelandic road authority’s summer traffic period; it is an operational definition, not a claim that weather is uniform across those months. The IMO also uses June–September when summarising the Icelandic summer (Veðurstofa Íslands, 2025). VHDU is not calculated as $ADU - SDU - VDU$. Annual traffic implied by ADU is first converted to a calendar-year total; the day-weighted SDU and VDU totals are subtracted, and the remainder is divided by the number of spring and autumn days:

$$VHDU = \frac{D_{year}ADU - D_{VDU}VDU - D_{SDU}SDU}{D_{VHDU}}.$$

Here, each D is the number of calendar days in that period. VHDU is therefore a residual estimate rather than an additional official traffic measure.

Daily traffic counts extracted from Vegagerðin reports cover 2019–2024 and provide observed totals at selected counter sites. They support both the traffic-response comparison and secondary daily-counter accident-rate analyses, but do not provide hourly vehicle-kilometres at every accident location. More than one counter may occur on a road section, and counter positions are estimated from the registered road geometry. These data remain separate from the primary O/E calculation.

Of 68,946 annual road-section–year–period records, 61,442 have nearby usable wind data; separately, 763,171 of 774,274 daily counter-days have a matched daytime wind summary. The more restrictive accident samples used by the traffic models are reported in Chapter 4.

3.1.5 Selection Summary

Table 3.7 summarises what was removed from each dataset or analysis sample. Selection for the study area and injury outcome is distinguished from invalid or missing measurements: an excluded record is not necessarily an erroneous record. The strict daily sample starts with the 1,863 rural injury accidents in 2019–2024; the other accident samples cover 2007–2025. Traffic periods count road-section–year–period combinations, whereas daily traffic counts counter-days. Zero daily traffic is retained in the descriptive traffic sample but excluded from positive-VKT analyses.

Table 3.7: Simple overview of data selection. Rows are separate selections, not successive steps; the last three count accidents.

Dataset / selection	Before	Removed	Retained	Reason
Accidents	126,607	120,193	6,414	Invalid time/location, urban, or no injury.
Weather observations	232,459,562	2,000,612	230,458,950	Missing, implausible, inconsistent or frozen wind.
Annual traffic periods	68,946	9,072	59,874	No usable weather or nonpositive traffic/length.
Daily counter-days	774,274	11,103	763,171	No usable daytime wind summary.
Primary weather sample	6,414	155	6,259	No weather match within 20 km and five minutes.
Annual traffic sample	6,414	1,289	5,125	Road linkage, exposure or weather eligibility.
Strict daily sample	1,863	1,250	613	Counter linkage, matching, traffic or coverage eligibility.

3.1.6 Descriptive Accident Patterns

Figure 3.2 shows when the selected accidents occurred and the conditions recorded with them. The blue and red groups are the same separate severity groups used in the O/E figures. These are raw counts: more accidents in an hour or season may reflect more traffic, and the panels do not estimate accident risk. Vehicle counts and accident-type composition are shown in Appendix A.1.

3.2 Data Preparation and Reproducibility

Table 3.8 summarises the sequence from source data to the reported results. Accident, weather and traffic records first undergo quality checks and selection for the study population. Qualifying accidents are then matched to nearby weather observations, while local weather frequencies provide the background for the primary O/E comparison. The matched-time analysis compares each accident with non-accident times under the same recurring calendar conditions. Finally, the available road-section and daily-counter data provide supporting traffic checks in their linked samples. This order keeps the primary weather comparison separate from the additional assumptions needed to estimate traffic exposure.

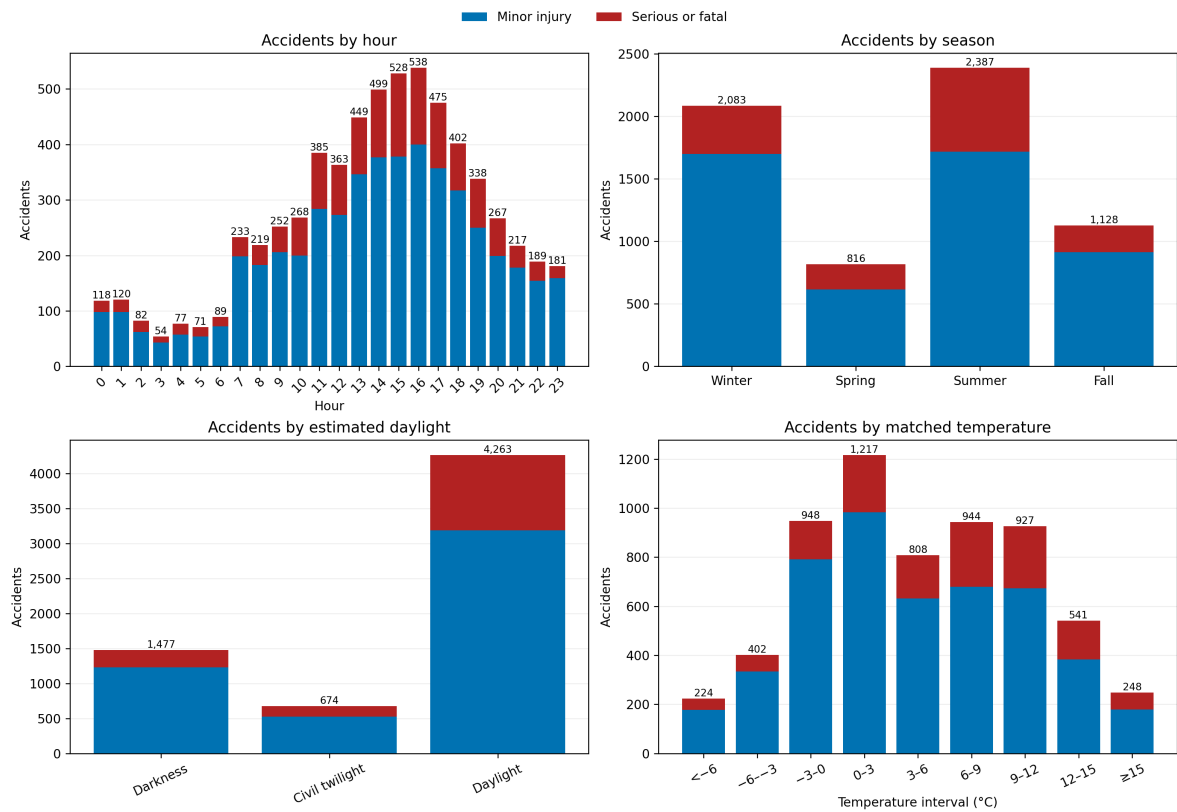


Figure 3.2: Rural injury accidents by hour, study season, estimated daylight and matched temperature. Blue shows minor injury (code 3), and red shows serious or fatal injury (codes 1–2). Labels give total accidents. Hour, season and daylight use 6,414 accidents; temperature uses 6,259 qualifying matches. These are descriptive counts, without traffic adjustment.

Table 3.8: Data preparation and analysis workflow

Stage	Main data	Output	Role in thesis
Data preparation	Accident, weather, road and traffic data	Quality-controlled datasets	Defines the study data.
Accident selection	Accident data and urban boundaries	Rural injury accidents	Defines the study population.
Weather matching	Accidents, weather observations and stations	Accident weather, local frequencies and control times	Creates the weather comparisons.
Weather-frequency O/E	Accident weather and local frequencies	Observed and expected counts	Primary analysis.
Matched-time analysis	Accident and control times	Matched odds ratios	Time-matched supporting comparison.
Traffic analyses	Accident weather, traffic and road lengths	Traffic-linked rates	Supporting analyses in smaller samples.

3.3 Analysis

3.3.1 Weather-Frequency Comparison

Accident-time mean wind speed f is the main weather measure and is divided into 5 m/s intervals. It is the 10-minute mean wind observation matched within five minutes of the accident time, not an instantaneous measurement. Wind gust f_g from the same observation and independently matched temperature are additional weather measures. For each measure, an accident is compared with the distribution at its matched station in the same study season, pooled across all study years. Expected counts are based on how often each condition occurred locally rather than on a national weather distribution. For example, if 5% of the relevant station–season observations fall in one wind interval, approximately 5% of that group’s accidents would be expected there under weather-frequency-proportional occurrence.

Let g identify a weather station and study season. Let A_{jg} be the number of accidents in weather interval j and group g , and let $A_g = \sum_j A_{jg}$. From all quality-controlled 10-minute observations, let T_{jg} be the number in interval j and T_g the total in group g . The local background frequency is

$$p_{jg} = \frac{T_{jg}}{T_g}.$$

If accident timing followed local wind frequency, the expected number in the interval would be $A_g p_{jg}$. The pooled observed/expected ratio is

$$R_j = \frac{\sum_g A_{jg}}{\sum_g A_g p_{jg}}.$$

A value of one means that the accident share equals the local time share of the weather interval. A value above one means that accidents are over-represented in that interval. Standardisation

within station and season prevents locations or seasons with different weather distributions from dominating the result.

The figures show the calculation separately for minor-injury accidents ($meiðsli = 3$) and serious-or-fatal accidents ($meiðsli \leq 2$). These groups do not overlap. Each group's expected count uses its own accident totals within station and season. The combined all-injury O/E is also retained for comparison with the other analyses; it is calculated from summed observed and expected counts, not by averaging the two group ratios. The O/E values are descriptive and are therefore shown without confidence intervals. Wind gust and temperature are handled in the same way as mean wind. Temperature is matched independently, and its intervals are below -6 , -6 to -3 , -3 to 0 , 0 to 3 , 3 to 6 , 6 to 9 , 9 to 12 , and at least 12°C . The O/E upper interval is combined at 12°C to keep the seasonal panels readable. The matched-time and annual-traffic models retain separate 12 – 15 and at-least- 15°C intervals. Estimates across those different upper intervals are not treated as direct numerical comparisons.

3.3.2 Matched-Time Comparison

This is a time-stratified case-crossover design: each accident forms one matched set. The accident time is the case time, and the control times are the other occurrences of the same weekday and clock time in the same calendar month and year. For example, a Tuesday accident at 14:20 is compared with the other Tuesdays at 14:20 in that month. Mean-wind and gust controls use the accident's matched wind station; temperature controls use its independently matched temperature station. Control weather must satisfy the same five-minute observation-time rule, and both station matches are restricted to 20 km. Conditional logistic regression compares weather at the accident time with weather at its matched control times. This time-stratified referent strategy was selected to avoid overlap bias and to control calendar time without selecting controls from outside the accident month (Janes et al., 2005).

The model asks whether weather at an accident time differs from weather at comparable non-accident times. Matching controls exactly for station, year, month, weekday, and clock time, but not for unusual traffic, road conditions, or other factors that change on a particular day. Continuous effects are reported per 5 m/s or 5°C together with categorical estimates. The mean-wind O/E remains the primary result.

Mean wind and temperature are also entered together as categorical variables to assess whether the wind estimate changes after including temperature at the same case and control times. Daylight is calculated at the accident location for the accident date and the same matched dates. Because month, weekday, and clock time are held fixed, only strata that change between darkness, civil twilight, and daylight contribute information.

Seasonal differences in the mean-wind association are tested in a separate matched-time model. Mean wind is grouped as 0–10, 10–15, and at least 15 m/s for stability. The model includes interactions between wind interval and season, and a likelihood-ratio test compares it with a model having one common wind association across seasons. The omnibus test is used instead of judging seasonal differences from separate confidence intervals.

A separate logistic model describes serious-or-fatal rather than minor injury among accidents with complete wind and temperature matches. It includes broad categories of mean wind, temperature, daylight, time of day, and season. The result concerns injury severity among recorded accidents. It is not a model of whether an accident occurs and does not replace the

traffic analyses.

3.3.3 Use of Traffic Data

The primary O/E denominator is the local frequency of wind conditions rather than traffic volume. Complete traffic data are unavailable at the required spatial and temporal resolution. O/E therefore does not account for fewer journeys on windy days. If traffic falls in strong wind while other factors remain unchanged, the accident-rate contrast per vehicle-kilometre would be larger than the O/E contrast. The traffic analyses examine this possibility. The descriptive estimated-rate calculation uses annual traffic on every eligible road section, including sections without an accident. The main conditional model has a narrower purpose and sample: its analysis retains 5,125 accidents in road-section–year–traffic-period strata that contain an accident and have a valid weather and traffic link. Both calculations use SDU and VDU where they are published, and the residual VH DU estimate for April–May and October–November.

Adding these traffic data to the primary O/E calculation would replace the broad study population with a smaller linked sample and combine seasonal traffic averages with 10-minute weather. O/E is therefore interpreted as relative accident occurrence, not accident rate per vehicle.

3.3.4 Road-Section Traffic Comparison

Traffic is included when accidents can be linked by road section, year, official traffic period, and wind interval k, j , estimated vehicle-kilometres are

$$V_{kj} = q_k L_k D_k p_{kj},$$

where q_k is SDU, VDU, or derived VH DU vehicles per day, L_k is section length, D_k is the number of calendar days, and p_{kj} is the local share of quality-controlled 10-minute observations in the wind interval. The accident rate is

$$\text{Rate}_j = \frac{A_j}{\sum_k V_{kj}} \times 100,000,000.$$

This method accounts for road length and for spatial and seasonal differences in published traffic. It still assumes that traffic within a traffic period is distributed across wind intervals in proportion to time. It does not observe traffic at the accident minute. The station nearest the road-section midpoint with qualifying data supplies p_{kj} . Each accident is matched to that road-section station rather than the accident-nearest station used in the primary O/E analysis. An accident is included only when the road-section station is within 20 km and has a valid observation within five minutes. The accident count and its estimated denominator therefore use the same weather station.

The main traffic result is a conditional Poisson model. Each road section, year, and traffic period has its own intercept. For stratum k and wind interval j , the model is

$$\log E(A_{kj}) = \alpha_k + \beta_j + \log V_{kj},$$

where A_{kj} is the accident count and $\log V_{kj}$ is the offset. The exponentiated coefficient $\exp(\beta_j)$ is the estimated rate ratio relative to 0–5 m/s within otherwise identical road-section-year-period strata.

The model is fitted separately for all injury accidents and for serious or fatal accidents. A seasonal extension uses year-specific station frequencies for winter, spring, summer, and autumn. Because individual 5 m/s upper bins are sparse within seasons, that extension uses 0–10, 10–15, and at least 15 m/s. Winter and summer use published VDU and SDU, respectively; spring and autumn use the same derived VHDU daily volume but retain separate weather frequencies and accident counts.

The descriptive rate and conditional model include different road records. The descriptive rate sums estimated vehicle-kilometres over every eligible road section. The conditional model includes only road-section-year-period strata containing at least one accident, because all-zero strata provide no within-stratum information: after conditioning on a stratum's total accident count, a total of zero contributes no likelihood for comparing its wind intervals. Those roads remain in the separate descriptive rate that sums all eligible exposure. Wind intervals with positive estimated vehicle-kilometres are kept within those groups. The two traffic results answer different questions and are not expected to reproduce the same rate ratios.

3.3.5 Daily Traffic Comparison

Daily traffic is analysed separately using daily weather summaries. Directional channels are first summed to one 24-hour count for each physical counter and date. The unit is then one fixed counter on one date; separate counter sites on the same road section are not summed. For each counter, year, month, and weekday, expected traffic is the arithmetic mean of the available daily counts in that same group. The traffic index is

$$I_{cd} = 100 \times \frac{Q_{cd}}{\text{mean}(Q_{c, \text{same year, month, weekday}})}.$$

An index below 100 indicates fewer counted vehicles than the mean for that local calendar context. Results are aggregated across counter-days and describe the participating sites rather than national traffic. The 24-hour count is not converted into hourly traffic. Mean wind is calculated from quality-controlled observations between 10:00 and 21:59, so this is a daily association rather than an estimate of traffic in that window. The daily-counter dataset contains 5,432 counter-days with a reported count of zero. They are kept because the reports do not distinguish a genuine zero or road closure from counter downtime; the appendix reports a check excluding these records. For each located counter, the nearest station within 20 km having usable observations that day is selected.

Sustained wind is described by the number of valid 10-minute observations in each counter-day with mean wind of at least 15 m/s; six observations equal one hour. Days must contain at least 108 valid observations. Traffic is compared across 0, more than 0–2, 2–6, and at least 6 hours of strong wind using the same counter–calendar standardisation.

The allocated daily-counter rate model uses a stricter accident linkage. Each 2019–2024 accident requires a counter on the exact registered road section and year within 20 km, positive

traffic on its date, and qualifying accident-time mean wind. For counter (c), date (d), and wind interval (j), the observed 24-hour count is allocated according to the counter station's valid 10-minute observations:

$$\hat{Q}_{cdj} = Q_{cd} \frac{N_{cdj}}{N_{cd}}.$$

Accidents are classified by accident-time f from the same station used for the counter-day denominator. That station must be within 20 km of both the accident and counter, and the accident observation must be within five minutes of the recorded time. Estimated traffic and accidents are aggregated by counter, year, and wind interval, and a conditional Poisson model gives each counter-year its own intercept. The allocation assumes uniform traffic across valid observation times; it is an estimated within-day traffic split, not observed hourly traffic. The main allocation uses all sufficiently complete 24-hour weather records.

A second daily-counter calculation uses the counter-section construction. For each road section and day, the observed daily vehicle count is multiplied by the road length assigned to that counter. This daily vehicle-kilometre total is allocated among weather intervals using the actual 10-minute observations at the assigned station from 07:00 to midnight on that same date. Accidents outside this time window are omitted. At least 92 of the expected 102 observations are required. For day i , counter-section j , and interval b ,

$$\hat{V}_{ijb} = Q_{ij} L_j \frac{N_{ijb}}{N_{ij}},$$

where Q_{ij} is the observed 24-hour count, L_j is the assigned section length, N_{ijb} is the number of valid same-day observations in interval b , and N_{ij} is the number of valid observations that day. The accident rate is the observed accidents divided by summed $\hat{V}_{ijb}$. This compares accidents per estimated vehicle-kilometre, but the weather-bin allocation is estimated: hourly traffic is not observed. Unlike the annual method, observed traffic on a windy date contributes its own daily total, so lower travel on that date is reflected in the denominator.

4 Results

Results follow the three research questions: weather-frequency O/E, matched-time comparisons, and traffic analyses in smaller linked samples. The main question is whether accidents are over-represented in strong wind relative to its local frequency. The subsequent comparisons assess whether that pattern persists under different time controls and exposure assumptions. Agreement is assessed by the pattern and uncertainty, not by treating O/E, odds ratios and estimated traffic-based rates as equivalent quantities.

4.1 Study Sample and Coverage

Table 4.1 shows how the analysis samples narrow as additional linkages are required. Most rural injury accidents enter the primary weather comparison, whereas traffic-linked analyses retain smaller groups. This difference matters when interpreting agreement between methods: the traffic results describe the roads and years with usable exposure information, not an identical sample with only its denominator changed. Figure 4.1 shows the accident counts and weather-match coverage by year, separating the size of the recorded sample from the proportion with qualifying weather data.

Table 4.1: Accident selection and analysis sample sizes.

Stage	Accidents	Purpose
Source records	126,607	Accident database, 2007–2025.
Rural injury accidents	6,414	Study population.
Primary O/E	6,259	Main weather analysis.
Matched-time	6,257	Time-matched comparison.
Annual traffic	5,125	Broader traffic analysis.
Daily-counter	762	Observed daily traffic totals.
Same-day vehicle-kilometres	613	Stricter counter-section analysis.

The 20 km threshold includes most rural injury accidents while bounding the distance to the weather proxy. Match coverage alone does not establish that station conditions equal those at the road: local terrain can alter wind over short distances. The road-section analysis is narrower than the main analysis, so its conclusions apply to a more geographically selective sample.

Wind and temperature are matched independently, but in the final analysis dataset both procedures retain the same 6,259 accidents and select the same station and observation time for each of them. Within 2019–2024, 762 of 1,863 accidents enter the allocated daily-rate model after requiring the exact road section, a valid observed counter-day, the same weather station for the accident and denominator, and all 20 km distance limits. The broader daily-counter selection step includes 767 accidents and is documented in the appendix.

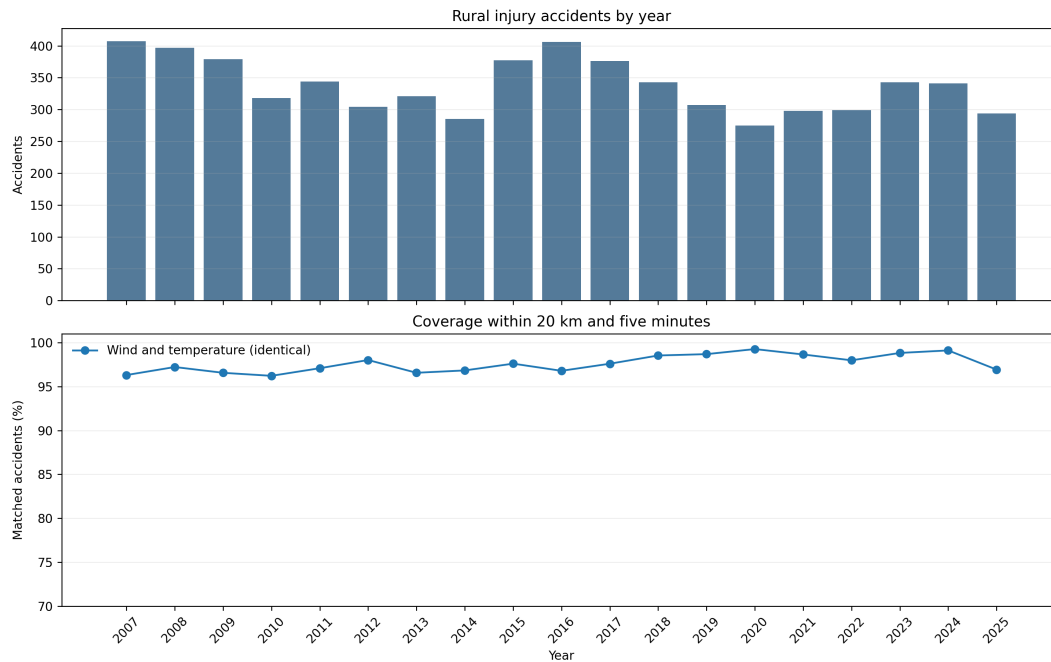


Figure 4.1: Annual rural injury-accident counts and the percentage matched to wind or temperature within 20 km and five minutes. The wind and temperature match counts are identical in every year; no missing temperature value is imputed.

4.2 Accident Frequency by Weather

Figures 4.2–4.4 show mean wind, gust and temperature for the full year and the four seasons. Blue bars show minor-injury accidents and red bars show serious-or-fatal accidents. The groups do not overlap. Counts are printed above the bars. Seasonal estimates in the upper weather intervals are descriptive because several contain few accidents. The grey horizontal line marks $O/E = 1$: the observed number of accidents equals the number expected from local weather frequency for that group. A bar above the line indicates over-representation relative to that frequency, not a measured increase in risk per vehicle.

4.2.1 Mean Wind

Figure 4.2 shows the main mean-wind pattern. The all-year ratios are close to one at low and moderate wind speeds and rise from the 15–20 m/s interval. Both severity groups show upper-wind over-representation when all seasons are pooled. The blue and red bars must nevertheless be read as separate descriptive comparisons: their heights alone do not test whether injury severity differs between wind categories.

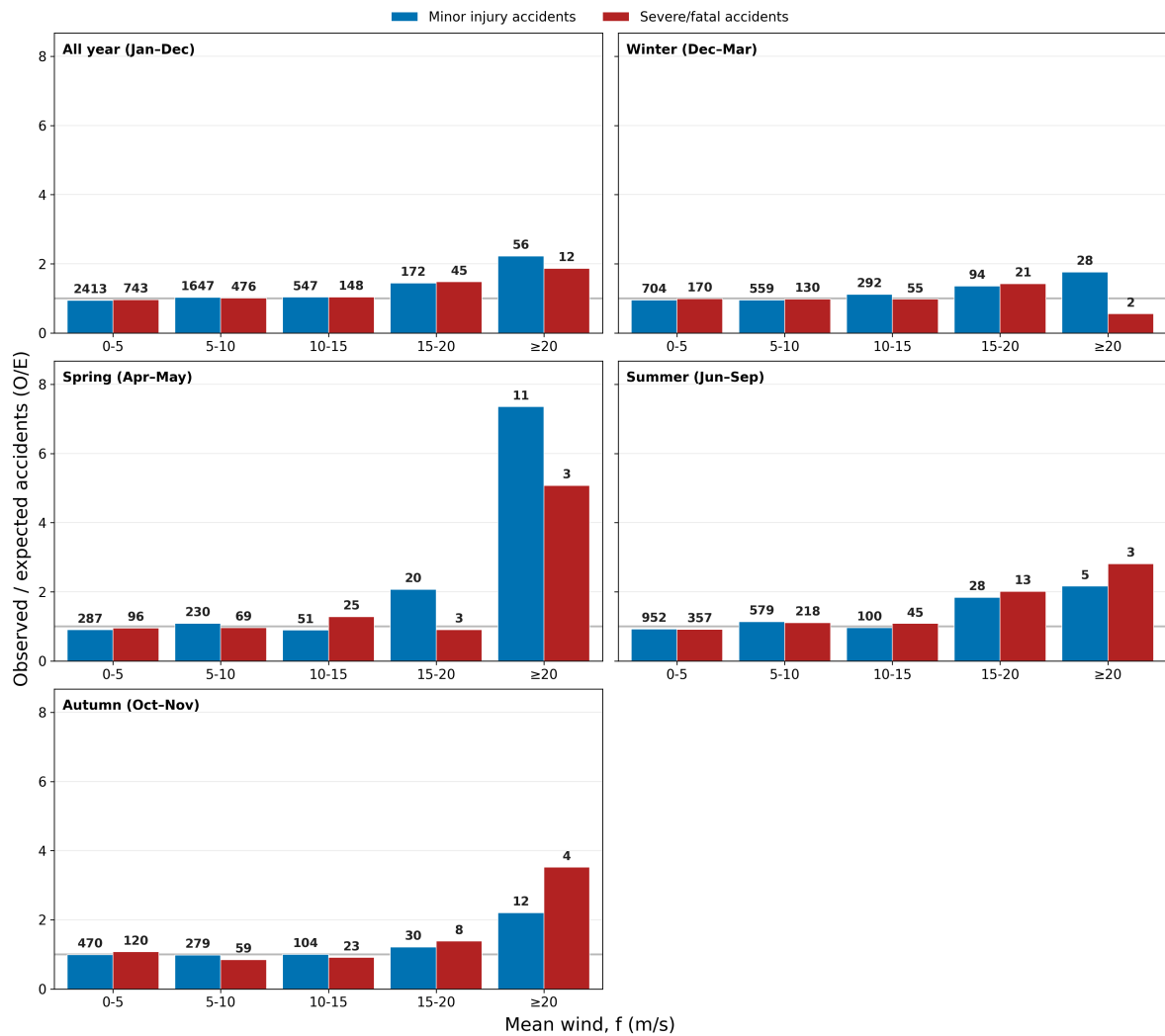


Figure 4.2: All-year and seasonal mean-wind O/E, 2007–2025. Blue bars show minor injury (*meiðsli* = 3); red bars show serious or fatal injury (*meiðsli* ≤ 2). Expected counts use local weather frequency separately for each group; numbers above bars are observed accidents.

At 20 m/s or above, the minor-injury group contains 56 accidents compared with 25.2 expected (O/E 2.22). The serious-or-fatal group contains 12 accidents compared with 6.4 expected (O/E 1.87). These are descriptive ratios; the smaller serious-or-fatal count does not support a precise statement about whether severity changes with wind. Combining both groups gives the retained all-injury result: 68 observed, 31.6 expected and O/E 2.15.

The seasonal panels show why the pooled pattern should not be interpreted as a uniform increase in every group and season. The tallest spring bars are based on few accidents, and the winter serious-or-fatal bar in the highest interval falls below one with only two observed accidents. Such small cells can produce large changes in O/E from a small change in counts. The figure supports an overall upper-wind pattern, but does not establish which season or injury group has the strongest association.

The 5 m/s grouping is used throughout because it gives a direct, readable description of the upper wind intervals without implying that the boundaries are physical thresholds.

4.2.2 Wind Gust

Accident-time wind gust is an additional weather measure rather than a daily maximum. Figure 4.3 shows that over-representation is concentrated towards the upper gust intervals. At 30 m/s or above, 52 minor-injury accidents were observed compared with 14.6 expected (O/E 3.56). The serious-or-fatal group contained nine accidents compared with 3.6 expected (O/E 2.47). Gust and mean wind are related measures and should not be treated as independent findings. Across both groups, the all-injury result remains 61 observed, 18.2 expected and O/E 3.34.

The seasonal upper bars are less stable than the pooled result because splitting the sample leaves few accidents in several cells. A tall seasonal bar therefore needs to be read alongside its count. Gust is taken from the same weather observation as mean wind, so agreement between the two figures is complementary evidence about windy conditions rather than an independent replication of the association.

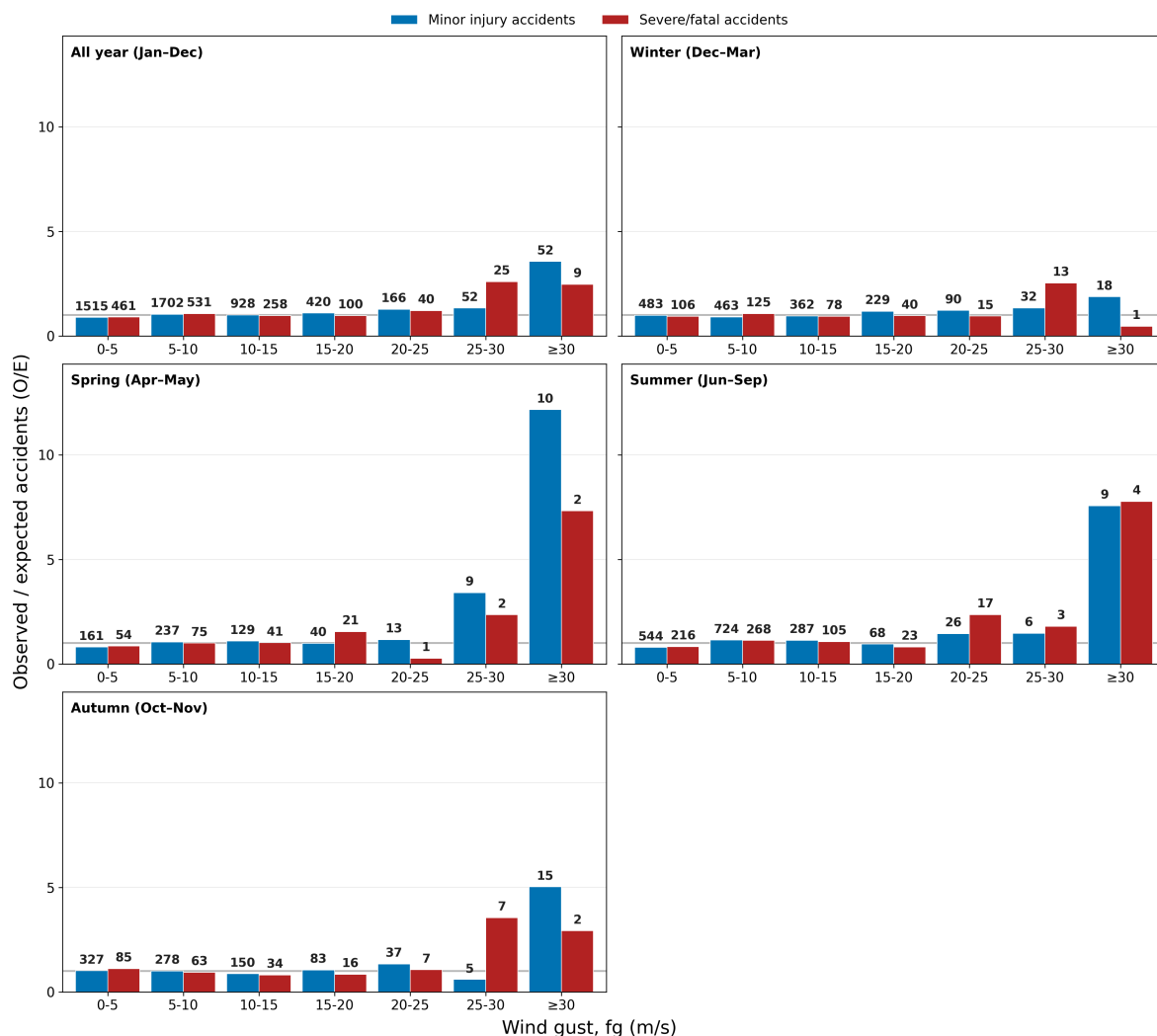


Figure 4.3: All-year and seasonal wind-gust O/E. Gust is taken from the 10-minute observation matched to the accident time; it is not a daily maximum. Blue denotes minor injury and red denotes serious or fatal injury; the groups do not overlap. Numbers above bars are observed accidents.

4.2.3 Temperature

Figure 4.4 shows a non-monotonic temperature pattern, which requires a different interpretation from the upper-wind elevation. Valid temperature was available for 6,259 accidents (97.58%). Coverage by year is shown in Figure 4.1, and no temperature value is imputed. After adjustment by temperature station and season, the estimates were non-monotonic. O/E was below one at 3–6°C for minor-injury accidents (632 observed and 881.7 expected; O/E 0.72), but above one at -3–0°C. At 12°C or above, 563 minor-injury accidents were observed compared with 362.0 expected (O/E 1.56). This pattern may reflect travel behaviour or weather and road conditions that vary with temperature. It is not interpreted as a causal temperature effect. For the serious-or-fatal group, all-year O/E is 0.69 at 3–6°C and 1.71 at 12°C or above (176 and 226 accidents, respectively).

The pooled curve should not be read as a single temperature gradient: temperature intervals occur with different frequencies across seasons, and the rare seasonal extremes provide limited information. Standardising within station and season accounts for those local frequencies, but does not control traffic volume, road-surface condition or other circumstances that vary within a season. The matched-time comparison below is therefore particularly important for assessing whether the warm-end pattern persists.

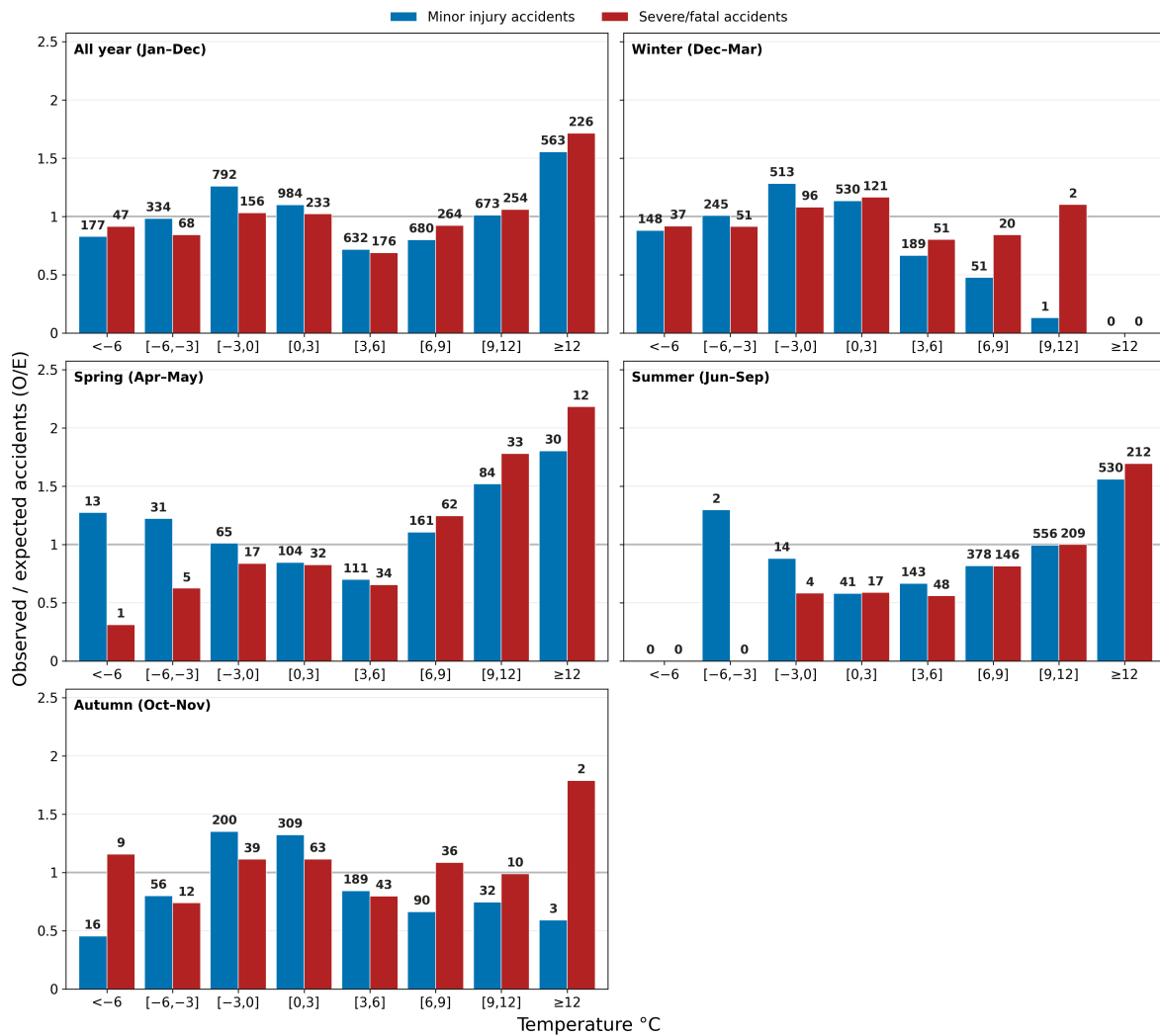


Figure 4.4: All-year and seasonal temperature O/E, standardised by matched temperature station and study season. Blue shows minor injury and red shows serious or fatal injury, with no overlap; numbers above bars are observed accidents. The non-monotonic pattern is interpreted cautiously.

Adding year to the station–season standardisation changes the selected estimates by no more than 0.02: for example, mean-wind O/E at 20 m/s or above changes from 2.15 to 2.17. The pre-specified station–season result is retained.

4.3 Matched-Time Results

The case-crossover analysis includes 6,257 accident strata for each weather measure. Table 4.2 collects the main categorical contrasts, leaving the text to explain their relationship to O/E. For categorical mean wind, the clearest increase occurs at 15 m/s or above: the odds ratio relative to 0–5 m/s is 1.61 (95% CI 1.38–1.87). This supports the high-wind pattern after exact control for station, year, month, weekday, and clock time. The lower and moderate wind contrasts are near one. The persistence of the upper-wind pattern under these controls makes an explanation based only on recurring calendar differences less plausible, while leaving

unusual traffic or road conditions on a particular day uncontrolled.

Table 4.2: Selected matched-time contrasts. Each model uses 6,257 accident strata; the joint wind contrast includes temperature. Intervals are 95% confidence intervals.

Comparison	OR	95% CI
Mean wind: ≥ 15 vs 0–5 m/s	1.61	1.38–1.87
Mean wind, with temperature: ≥ 15 vs 0–5 m/s	1.68	1.44–1.96
Gust: ≥ 30 vs 0–10 m/s	2.74	1.96–3.85
Temperature: $-3-0$ vs $0-3$ °C	1.13	1.02–1.25
Temperature: $3-6$ vs $0-3$ °C	0.59	0.53–0.65
Temperature: ≥ 15 vs $0-3$ °C	0.99	0.80–1.23

For gust, the categorical estimates rise across the upper intervals and agree with the upper-gust O/E pattern. The largest contrast is less precise than those based on more common conditions. Because mean wind and gust describe related aspects of the same observation, agreement does not separate their individual contributions.

The temperature estimates are not monotonic. Relative to $0-3$ °C, the estimate is somewhat higher just below freezing, lower at $3-6$ °C, and near one at 15 °C or above. This last interval is narrower than the O/E at-least- 12 °C interval. The matched-time result reproduces the dip at moderate temperatures but not the warm-tail increase in the O/E analysis. The methods therefore give different results at warm temperatures, which calls for caution.

When mean wind and temperature are included together, the matched odds ratio for mean wind of at least 15 versus 0–5 m/s is 1.68 (95% CI 1.44–1.96). This is close to the single-variable estimate of 1.61, and the non-monotonic temperature pattern remains.

4.4 Daylight, Season, and Severity

The matched-date daylight comparison holds location, month, weekday, and clock time fixed. Relative to daylight, the odds ratio is 1.14 (95% CI 0.88–1.49) for darkness and 1.10 (0.92–1.31) for civil twilight. Only 985 of the 6,414 strata change daylight class within their matched dates. The result provides no clear evidence of a daylight association and is much less informative than the wind analysis; it does not use vehicle counts. Most matched sets remain in the same daylight category and therefore cannot distinguish the categories. This limited within-set variation, together with intervals spanning one, is a reason for caution rather than evidence that daylight is irrelevant to road safety.

Among the 6,259 accidents in the adjusted severity model, 1,424 are serious or fatal. Mean wind of at least 15 m/s has an odds ratio of 0.93 (95% CI 0.69–1.26) relative to 0–10 m/s. Darkness has an odds ratio of 0.89 (0.71–1.12) relative to daylight. Thus the data show a clear strong-wind pattern for accident occurrence, but not for severity conditional on a recorded injury accident. The complete adjusted estimates are shown in Table A.1 in the appendix.

The matched-time wind-by-season comparison uses the same 6,257 mean-wind strata as the case-crossover analysis, with coarse intervals for stability. The omnibus interaction test gives $\chi^2(6) = 14.30$, $p = 0.026$. The estimated odds ratios for at least 15 versus 0–10 m/s

are 1.44 in winter, 2.02 in spring, 2.53 in summer, and 1.37 in autumn. The omnibus test provides evidence that at least some seasonal wind associations differ, but it does not identify a particular pair of seasons as different. Season-specific estimates are therefore not ranked.

4.5 Traffic Results

The annual analysis has broader coverage, whereas the daily-counter analysis uses observed daily totals at fewer locations. Both require traffic to be allocated among wind intervals; neither observes traffic at the accident minute. Table 4.3 contrasts the annual and strict daily-counter exposure constructions and their main limitations.

Table 4.3: Traffic inputs for the supporting vehicle-kilometre analyses.

Analysis	Years (accidents)	Traffic basis	Main limitation
Annual road sections	2007–2025 (5,125)	VDU, SDU and derived VHDU	Traffic is allocated by wind frequency within each period.
Daily counter sections	2019–2024 (613)	Observed daily totals	Within-day traffic is estimated; locations and upper bins are sparse.

4.5.1 Broader Annual-Traffic Comparison

Table 4.4 reports descriptive accident rates based on estimated vehicle-kilometres. The rates rise across the wind intervals, but these pooled values also reflect which road sections and periods contribute exposure to each interval. The conditional comparison described below instead compares intervals within road section, year and traffic period.

Table 4.4: Estimated rural injury-accident rate by mean wind speed, all traffic periods

Mean wind	Accidents	Estimated accidents per 100 million vehicle-km
0–5	2,539	9.0
5–10	1,755	12.8
10–15	584	16.7
15–20	186	30.0
20–25	44	46.3
≥25	17	98.9

The within-stratum model also shows higher rates in the upper wind intervals. Its estimate for 20–25 versus 0–5 m/s is 2.34. This conditional comparison differs from the pooled descriptive rates because it controls the comparison within road section, year and traffic period.

The annual analysis retains 5,125 accidents. The conditional Poisson fit uses 5,122 of these in 4,133 informative road-section-year-traffic-period groups. Relative to 0–5 m/s, the estimated within-stratum rate ratio is 1.67 (95% CI 1.44–1.95) at 15–20 m/s, 2.34 (1.72–3.17) at 20–25 m/s, and 5.16 (3.15–8.44) at 25 m/s or above. The uppermost estimate contains 17 accidents and is less precise. Traffic is allocated within each period according to local wind frequency.

The elevation from 15–20 m/s agrees in direction with both primary O/E and matched time. Its numerical size is not directly comparable with either: the annual model uses a selected road-linked sample, a different reference comparison and constructed exposure. The wider upper-bin intervals also mean that the largest point estimate should not carry the main conclusion.

The serious-or-fatal model includes 1,088 accidents. Relative to 0–5 m/s, its rate ratio is 2.04 (95% CI 1.49–2.79) at 15–20 m/s and 1.88 (0.88–4.02) at 20–25 m/s. The latter interval includes one and does not provide clear evidence of an elevated rate in that bin. The at-least-25 m/s estimate is 5.51 (1.97–15.40), but contains only four accidents and is correspondingly imprecise.

Using wider seasonal intervals, the estimated rate ratio for at least 15 versus 0–10 m/s is above one in all four seasons. These values are supporting checks only: spring and autumn use derived VHDU, and the upper seasonal cells are too small for a precise ranking.

4.5.2 Daily-Counter Checks

Observed daily traffic falls as strong wind persists. Relative to the calendar expectation at the same counter, traffic is 86.8% on days with at least six hours at $f \geq 15$ m/s. This shows that traffic volume is not constant across weather conditions at the selected counters. It does not quantify how traffic changed across the full primary accident sample, because the counters are geographically selective and do not provide complete hourly traffic at accident times. The finding motivates the exposure checks rather than supplying a numerical correction to primary O/E.

The allocated daily-counter model retains 762 accidents. Its rate ratio for at least 15 versus 0–10 m/s is 3.62 (95% CI 2.65–4.94), based on 47 accidents in the upper interval. A stricter counter-section calculation includes road length and actual same-day weather for 613 daytime accidents. Its upper mean-wind interval contains only ten accidents. Both calculations start from observed daily totals but estimate their within-day distribution, and both are limited to selected counter locations. They are therefore supporting checks, not the primary result.

The two daily calculations also ask different exposure questions: one fits relative rates using allocated counter totals, while the stricter calculation includes assigned road length and reports accidents per estimated VKT. Their agreement in direction is useful, but cannot remove the uncertainty in the within-day allocation or the sparse upper-wind counts.

Appendix Table A.5 addresses the two most important data questions. Restricting the rate model to the official VDU and SDU periods reduces the upper estimates, but they remain above one. Excluding daily counter records reported as zero changes the 20–25 m/s traffic estimate from 68.9% to 69.3%, so those records do not explain the observed decline.

Table 4.5 brings the primary result and supporting comparisons together. The common finding is an upper-wind association; the table also shows why each estimate retains a distinct interpretation.

Table 4.5: Main result and supporting comparisons.

Analysis	Estimate	Role
Weather-frequency O/E	≥ 20 m/s: O/E 2.15	Primary result; local wind frequency, no traffic.
Matched time	≥ 15 vs 0–5 m/s: OR 1.61 (1.38–1.87)	Same station and calendar time.
Annual traffic	20–25 vs 0–5 m/s: RR 2.34 (1.72–3.17)	Broader traffic sample; traffic allocated within periods.
Same-day traffic	≥ 20 vs 0–5 m/s: 88.5 vs 6.8 per 100 million VKT	Observed daily totals; 613 accidents, 10 in the upper interval.

The estimates measure different quantities and should not be compared by magnitude. Their direction is consistent, but hourly traffic is unavailable for the hour-of-day and daylight analyses.

5 Discussion

5.1 Interpretation

The primary O/E analysis shows that strong mean wind is over-represented at the times of rural injury accidents relative to its local frequency. The matched-time analysis supports this pattern after controlling for station and recurring calendar time. The traffic results point in the same direction but come from smaller samples and estimated exposure. The evidence therefore supports an association, not a causal effect or a national per-vehicle risk.

Observed daily traffic is lower when strong wind persists at the selected counters. This cautions against treating weather time as a direct measure of vehicle exposure, but cannot establish how traffic contributed to the primary O/E pattern across the wider network. Neither traffic source observes traffic at the accident minute: the annual analysis allocates seasonal traffic by wind frequency, and the daily analyses allocate observed totals within the day. These results support rather than replace the primary comparison.

Gust also has an upper-tail pattern, although it is closely related to mean wind. Temperature is non-monotonic and differs at the warm end between the O/E and matched-time analyses. The daylight comparison gives no clear association, and the severity model gives no clear evidence that strong wind increases severity among recorded injury accidents. Sparse seasonal upper intervals do not support ranking seasons or outcome groups.

The main result agrees with earlier studies comparing accident shares with the time spent in adverse weather (Edwards, 1994) and with the engineering evidence that crosswind can impair vehicle stability (Baker, 1986, 1994). This study adds a longer Icelandic series, local station-and-season standardisation, and matched-time and traffic checks; it does not establish a new physical wind threshold. The non-monotonic temperature estimates are consistent with previous evidence that temperature associations can vary by range and season (Lee et al., 2014; Zhan et al., 2020). The decrease in daily traffic during sustained wind is consistent with recorder-based evidence that adverse weather changes travel demand (Sathiaraj et al., 2018).

5.2 Strengths

The study combines a long accident series with 10-minute weather observations, local weather-frequency standardisation, and explicit reporting of match coverage and observed counts. The matched-time and traffic analyses address different weaknesses of the primary comparison. Documented transformations, exclusions, and automated checks make the reported results reproducible.

5.3 Limitations

- The main result is not traffic-adjusted. It should not be interpreted as accident risk per vehicle-kilometre.
- The case-crossover analysis controls recurring calendar time but not unusual traffic, road conditions, visibility, or closures on a particular day. Its ≥ 15 m/s category summarises the upper-wind pattern rather than a physical threshold.
- A weather station up to 20 km away is a proxy for conditions at the accident location. Terrain can produce important differences over shorter distances.
- The annual traffic analysis allocates seasonal traffic across wind intervals and does not observe traffic at the accident minute.
- Daily counters cover selected sites in 2019–2024 and provide no hourly traffic. Their totals are allocated within the day, and an assigned counter need not have been passed by the accident vehicle.
- Estimated rates are restricted to rural injury accidents on linked road sections. They should not be compared directly with broad national or international accident rates unless the outcome and vehicle-kilometre traffic quantity are defined in the same way.
- The highest wind intervals contain relatively few accidents. Several intervals and comparison results are shown, so individual estimates in the upper tail should be interpreted with caution.
- Fixed 2020–2024 urban boundaries are applied to the entire 2007–2025 accident period.
- The analysis is observational. Unmeasured road condition, vehicle type, speed, visibility, and driver behaviour can contribute to the pattern.

5.4 Future Work

The most valuable addition would be verified hourly traffic with counter coordinates, road-section identifiers, direction information, and documented downtime. Linked to the same weather stations, these data could distinguish vehicle classes and wind direction relative to the road. Daily totals cannot provide that information.

6 Conclusion

Of 6,414 rural injury accidents during 2007–2025, 6,259 were matched to valid weather observations within 20 km and five minutes. At mean wind speeds of at least 20 m/s, 68 injury accidents occurred compared with 31.6 expected from local weather frequency (O/E 2.15). The serious-or-fatal subset had O/E 1.87, but its smaller count limits comparison between severity groups.

The matched-time analysis supports the upper-wind pattern: the odds ratio for at least 15 m/s versus 0–5 m/s was 1.61. Gust gives a similar upper-tail pattern, whereas the non-monotonic temperature estimates differ between the O/E and matched-time analyses at warmer temperatures. The temperature finding therefore remains uncertain.

Annual and daily traffic analyses support the high-wind pattern, and observed daily traffic is lower during sustained strong wind. These analyses cover narrower, selected samples and estimate traffic within wind intervals from seasonal or daily totals; they do not provide complete hourly exposure at accident times.

The analyses therefore show that rural injury accidents are relatively more frequent during strong mean wind in Iceland. This is not a causal estimate or a complete national measure of accident risk per vehicle. Such a measure would require linked hourly traffic data with wider geographic coverage.

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A Supporting Checks

This appendix contains the checks most relevant to interpreting the main results. Additional checks are documented separately.

A.1 Accident Characteristics

These descriptive figures complement the hour, season and daylight counts in the data chapter. They describe the selected injury accidents, without a traffic-exposure denominator.

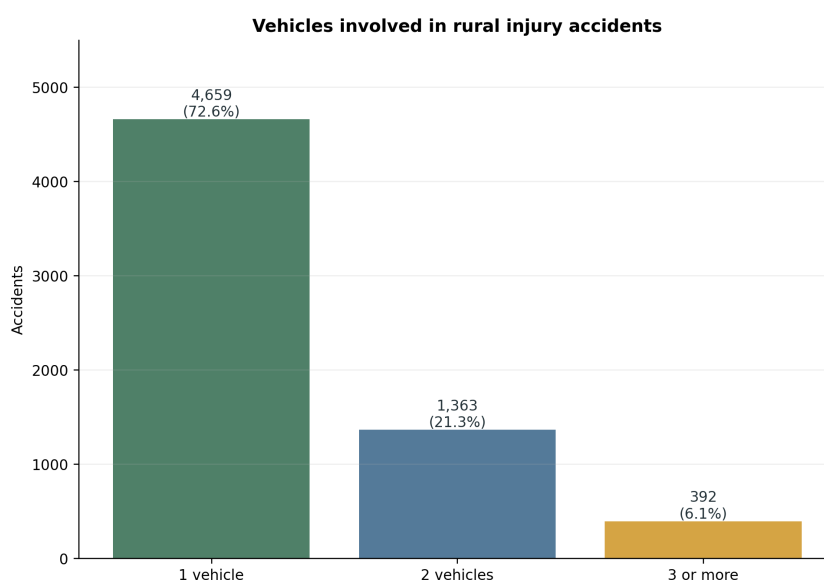


Figure A.1: Number of registered vehicles involved in the 6,414 rural injury accidents. Counts describe the selected sample, not risk by vehicle count.

A.2 Adjusted Severity Model

Table A.1 shows the adjusted comparison of serious-or-fatal and minor-injury accidents. It describes severity among recorded injury accidents; it does not estimate accident occurrence or risk per vehicle.

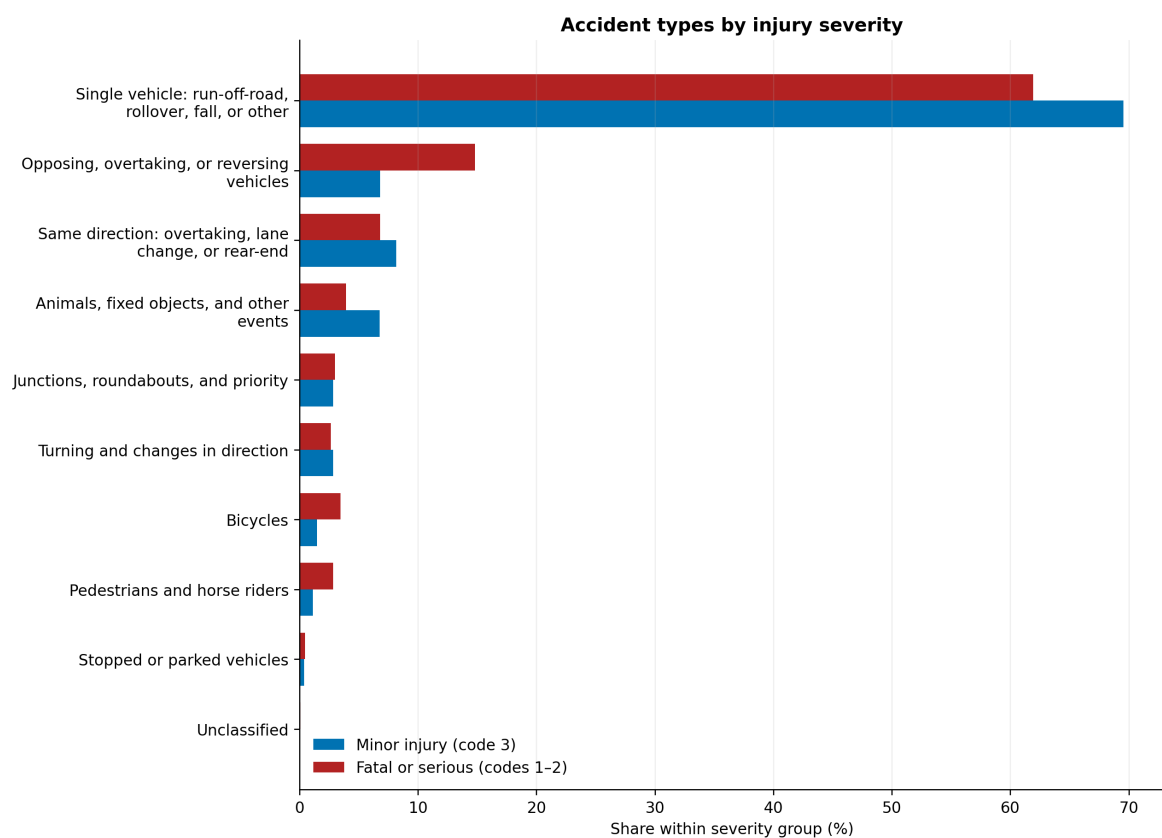


Figure A.2: Accident-type composition for minor injury (blue, code 3) and serious or fatal injury (red, codes 1–2). Percentages sum to 100 within each separate severity group.

Table A.1: Adjusted odds ratios for a serious-or-fatal outcome among recorded injury accidents. All listed variables are included in the same model.

Variable	Comparison	Adjusted OR (95% CI)
Mean wind	10-15 m/s vs 0-10 m/s	0.99 (0.82–1.21)
Mean wind	≥ 15 m/s vs 0-10 m/s	0.93 (0.69–1.26)
Temperature	< -6 °C vs 0-3 °C	1.13 (0.79–1.62)
Temperature	-6 – -3 °C vs 0-3 °C	0.85 (0.63–1.15)
Temperature	-3 – 0 °C vs 0-3 °C	0.83 (0.66–1.04)
Temperature	3 – 6 °C vs 0-3 °C	1.11 (0.88–1.39)
Temperature	6 – 9 °C vs 0-3 °C	1.41 (1.10–1.79)
Temperature	9 – 12 °C vs 0-3 °C	1.28 (0.98–1.68)
Temperature	12 – 15 °C vs 0-3 °C	1.33 (0.98–1.81)
Temperature	≥ 15 °C vs 0-3 °C	1.21 (0.83–1.76)
Daylight	Darkness vs Daylight	0.89 (0.71–1.12)
Daylight	Civil twilight vs Daylight	1.09 (0.87–1.36)
Time of day	00-06 vs 10-16	0.86 (0.64–1.16)
Time of day	06-10 vs 10-16	0.65 (0.52–0.82)
Time of day	16-20 vs 10-16	0.98 (0.84–1.14)
Time of day	20-24 vs 10-16	0.77 (0.61–0.98)
Season	Winter vs Summer	0.82 (0.64–1.05)
Season	Spring vs Summer	0.96 (0.78–1.18)
Season	Autumn vs Summer	0.83 (0.65–1.05)

A.3 Daily-Counter Sample

The following tables show the narrowing of the 2019–2024 accident sample and compare retained accidents with those excluded by the daily-counter linkage. The 767 records at the last broad selection step include five records that do not enter the final allocated-rate model; that model retains 762 accidents.

Table A.2: Selection of accidents for the observed daily-counter analysis

Selection step	Accidents	Share of 2019–2024 sample
Rural injury accidents in 2019–2024	1,863	100.0%
Exact road-section/year counter candidate	864	46.4%
Assigned counter within 20 km	863	46.3%
Positive count and valid full-day wind	767	41.2%

Table A.3: Characteristics of accidents retained and excluded by the daily-counter linkage

Characteristic	All	No link	Linked, excluded	Retained
Accidents	1,863	999	102	762
Share of all accidents	100.0%	53.6%	5.5%	40.9%
Distinct road sections	563	374	56	189
Serious or fatal	25.6%	29.4%	18.6%	21.5%
One vehicle	67.7%	75.2%	54.9%	59.7%
Winter	32.6%	28.0%	40.2%	37.5%
Spring	13.0%	11.7%	8.8%	15.4%
Summer	36.0%	41.8%	33.3%	28.6%
Autumn	18.4%	18.4%	17.6%	18.5%

The geographic selectivity of this traffic-linked sample is documented in Figure A.3.

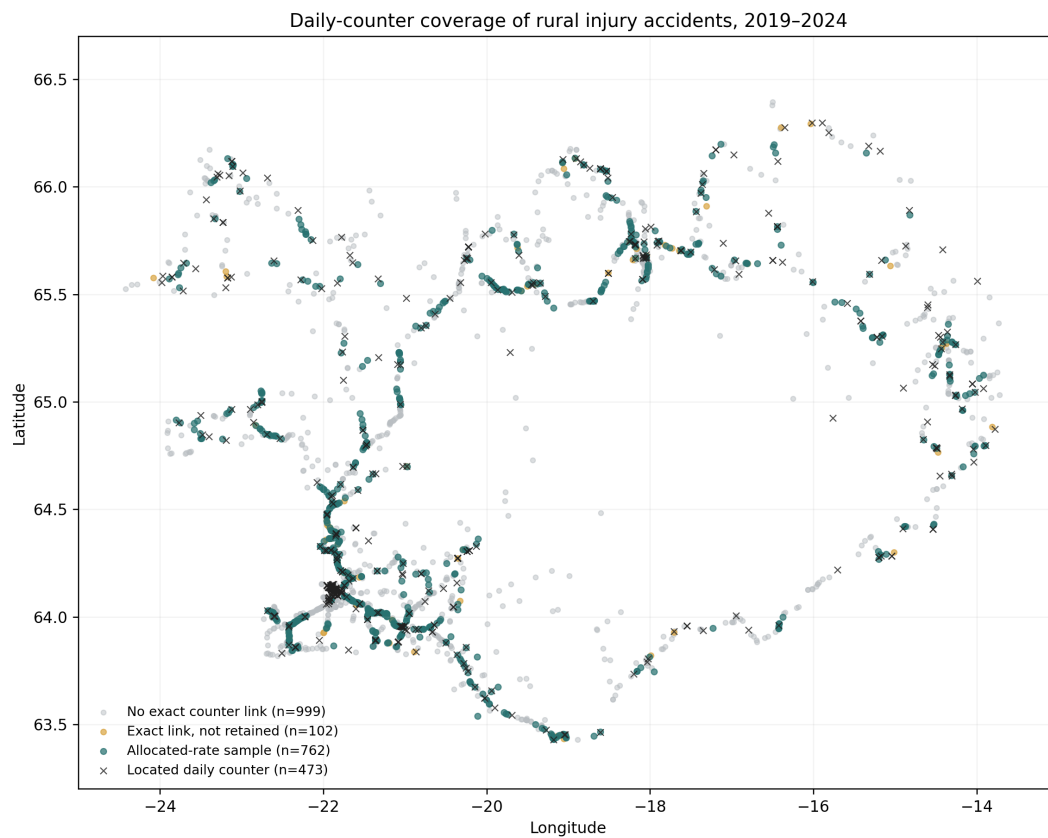


Figure A.3: Geographic coverage of located daily counters and rural injury accidents retained or excluded by the daily-counter linkage, 2019–2024.

A.4 Traffic Sensitivity Checks

The assignment-distance check shows how the broad daily-counter result changes under 5, 10, and 20 km limits. It is retained to make the spatial selection rule explicit.

Table A.4: Upper daily-counter estimate under three assignment distances

Maximum distance	Included accidents	Upper-bin accidents	Rate ratio	95% CI
5 km	585	12	2.91	1.58–5.37
10 km	750	14	2.78	1.58–4.88
20 km	767	14	2.73	1.55–4.78

Table A.5 combines the remaining traffic checks. It compares the annual model using all traffic periods with the version using only official VDU and SDU, reports the effect of excluding zero counter-days, and records exclusions caused by nonpositive VDU or derived spring/autumn daily traffic values.

Table A.5: Comparison and quality checks for the traffic analyses

Check	Data included	Estimate (95% interval)	Records
Rate model, 20-25 m/s	All VDU, SDU, and derived VHDU periods	RR 2.34 (1.72–3.17)	44 accidents
Rate model, 20-25 m/s	Official VDU and SDU periods only	RR 2.03 (1.38–2.96)	28 accidents
Rate model, ≥ 25 m/s	All VDU, SDU, and derived VHDU periods	RR 5.16 (3.15–8.44)	17 accidents
Rate model, ≥ 25 m/s	Official VDU and SDU periods only	RR 2.84 (1.34–6.04)	7 accidents
Daily traffic, 20-25 m/s	Retain zero counter-days	68.94% (62.33–74.09)	1,175 days
Daily traffic, 20-25 m/s	Exclude zero counter-days	69.29% (62.70–74.52)	1,090 days
Nonpositive published VDU	Excluded, not imputed	6.57% excluded	1,509 section-years
Nonpositive derived VHDU	Excluded, not imputed	2.40% excluded	552 section-years